

PRODUCT REQUIREMENTS DOCUMENT

Complaint Management Portal

Stack	React + Ant Design Django REST Framework PostgreSQL
Roles	Admin User
Date	June 16, 2026

1. Introduction & Overview

1.1 Purpose

A web application enabling end-users to lodge complaints and allowing administrators to track, review, and manage them.

1.2 Project Scope

- A web portal built with React 18+ and Ant Design 5.x on the frontend.
- A RESTful API backend built with Django 4.x, Django REST Framework (DRF), and PostgreSQL 15+.
- Two distinct user roles: Admin and User (complainant).
- Core workflows: complaint submission by users; complaint viewing and management by admins.
- JWT-based authentication for both roles.

2. User Stories & Acceptance Criteria

2.1 Admin Stories

ID	User Story	Acceptance Criteria
US-A01	As an Admin, I want to log in with my credentials so I can access the admin dashboard.	Login form accepts email + password. JWT returned on success. Redirect to dashboard on success. Error message on failure.
US-A02	As an Admin, I want to view a list of all complaints so I can get an overview of submitted issues.	Table shows ID, title, status, submitted date, and complainant. Paginated (20 per page). Sortable columns.
US-A03	As an Admin, I want to filter complaints by status, and date range so I can find relevant cases quickly.	Filter panel with Status, and Date Range controls. Filters are combinable. Reset button clears all filters.
US-A04	As an Admin, I want to view the full detail of a complaint so I can understand the issue.	Detail view shows all fields: title, description, status, timestamps, complainant info.
US-A05	As an Admin, I want to update the status of a complaint so I can track resolution progress.	Status dropdown: Open, In Progress, Resolved, Closed. Change is saved immediately with timestamp.

ID	User Story	Acceptance Criteria
US-A06	As an Admin, I want to add internal notes to a complaint so I can document actions taken.	Text area for notes. Notes are saved with author and timestamp. Only admins can see notes.
US-A07	As an Admin, I want to log out so my session is terminated securely.	Logout clears JWT from client. Subsequent API calls return 401.

2.2 User (Complainant) Stories

ID	User Story	Acceptance Criteria
US-U01	As a User, I want to access the complaint form without mandatory registration so I can quickly report an issue.	Complaint form is accessible at /complaints/new. No login required.
US-U02	As a User, I want to fill in my name, email, title, and description so my complaint is complete.	Form validates required fields. Email format validated. Description min 20 chars. Success message on submit.
US-U03	As a User, I want to receive a unique complaint reference number so I can track my submission.	Reference number shown on success screen. Unique alphanumeric ID (e.g., CMP-2024-0001).
US-U04	As a User, I want to view the status of my complaint using my reference number and email so I can follow progress.	Status check page at /complaints/track. Enter ref no. + email. Shows current status and timestamps.

3. Functional Requirements

3.1 Authentication Module

- Admin login via email + password. JWT access token (15 min) + refresh token (7 days) issued on success.
- Refresh token rotation: every refresh call returns a new pair.
- Token stored in HttpOnly cookie (refresh) and memory/localStorage (access).
- All admin API endpoints protected by IsAuthenticated + IsAdminUser permission class.
- Failed login attempts: lock account after 5 consecutive failures for 15 minutes.

3.2 Complaint Submission

- Public endpoint — no authentication required.
- Fields: Full Name (required), Email (required, validated), Phone (optional), Title (required, max 150 chars), Description (required, min 20 / max 2000 chars).
- On submission: complaint saved with status=Open, auto-generated reference number, created_at timestamp.
- Rate limiting: max 5 submissions per IP per hour to prevent spam.

3.3 Complaint Tracking (User-Facing)

- Users enter reference number + email to look up their complaint.
- Returns: status, submitted date, last updated date. Does NOT expose admin notes.

3.4 Admin Dashboard

- Protected by admin JWT. Displays summary cards: Total, Open, In Progress, Resolved, Closed counts.
- Complaints table with pagination (default 20/page, configurable to 50/100).
- Sort by: Date Submitted (default desc), Status.
- Filter by: Status (multi-select), Date Range (date picker).
- Search by: Reference Number or complainant name/email (full-text).

3.5 Complaint Detail & Management (Admin)

- Admin can view all complaint fields including complainant contact info.
- Status transitions allowed: Open → In Progress → Resolved → Closed. Cannot revert to Open from Closed.
- Internal notes: append-only log with admin name, note text, and timestamp.
- Audit trail: every status change logged with who changed it and when.

14. Non-Functional Requirements

Category	Requirement
Performance	API response time
Performance	Page load (Largest Contentful Paint)
Availability	Uptime
Security	Auth
Security	Data
Security	Input

Category	Requirement
Usability	Accessibility
Usability	Responsiveness
Maintainability	Code coverage
Compliance	Data retention

5. System Architecture

5.1 High-Level Architecture

Layer	Technology
Presentation	React 18 + Ant Design 5
API Gateway / Backend	Django 4 + DRF 3
Database	PostgreSQL 15

5.2 Directory Structure

Backend (Django)

Path	Purpose
complaint_portal/	Django project root (settings, urls, wsgi)
apps/accounts/	Custom User model, JWT auth views
apps/complaints/	Complaint model, views, serializers, URLs
apps/core/	Shared utilities, permissions, pagination
requirements.txt	Python dependencies
.env	Environment variables (never commit)

Frontend (React)

Path	Purpose
src/pages/	AdminLogin, Dashboard, ComplaintDetail, SubmitComplaint, TrackComplaint
src/components/	Reusable UI: ComplaintTable, StatusBadge, FilterPanel, StatCard

Path	Purpose
src/services/	Axios instance, API call functions (authAPI, complaintAPI)
src/store/	Redux slices: auth, complaints
src/routes/	ProtectedRoute HOC, route definitions
src/utils/	Token helpers, formatters, validators

6. Data Models

6.1 User (Admin)

Field	Type	Constraints	Notes
id	UUID	PK, auto-generated	Primary key
email	VARCHAR(254)	UNIQUE, NOT NULL	Used as login username
password	VARCHAR(255)	NOT NULL	bcrypt hashed
first_name	VARCHAR(100)	NOT NULL	
last_name	VARCHAR(100)	NOT NULL	
is_staff	BOOLEAN	DEFAULT TRUE	Grants admin access
is_active	BOOLEAN	DEFAULT TRUE	Soft disable accounts
last_login	TIMESTAMPTZ	NULLABLE	
date_joined	TIMESTAMPTZ	DEFAULT NOW()	

6.2 Complaint

Field	Type	Constraints	Notes
id	UUID	PK, auto-generated	
reference_number	VARCHAR(20)	UNIQUE, NOT NULL	e.g. CMP-2024-0001
complainant_name	VARCHAR(200)	NOT NULL	
complainant_email	VARCHAR(254)	NOT NULL	Used for tracking lookup
complainant_phone	VARCHAR(20)	NULLABLE	

Field	Type	Constraints	Notes
category	VARCHAR(50)	NOT NULL	Enum: see categories list
title	VARCHAR(150)	NOT NULL	
description	TEXT	NOT NULL	Min 20, max 2000 chars
status	VARCHAR(20)	DEFAULT 'open'	open in_progress resolved closed
created_at	TIMESTAMP TZ	DEFAULT NOW()	Auto-set on creation
updated_at	TIMESTAMP TZ	AUTO	Auto-updated on save
resolved_at	TIMESTAMP TZ	NULLABLE	Set when status → resolved

6.3 ComplaintNote (Admin Internal Notes)

Field	Type	Constraints	Notes
id	UUID	PK, auto-generated	
complaint	FK → Complaint	ON DELETE CASCADE	Parent complaint
author	FK → User	ON DELETE SET NULL	Admin who wrote the note
note_text	TEXT	NOT NULL	
created_at	TIMESTAMP TZ	DEFAULT NOW()	

6.4 AuditLog

Field	Type	Constraints	Notes
id	UUID	PK	
complaint	FK → Complaint	ON DELETE CASCADE	
changed_by	FK → User	NULLABLE	NULL = system
old_status	VARCHAR(20)	NOT NULL	
new_status	VARCHAR(20)	NOT NULL	
changed_at	TIMESTAMP TZ	DEFAULT NOW()	

7. API Endpoints

7.1 Authentication

Method	Endpoint	Auth	Request Body	Response	Notes
POST	/api/auth/login/	None	{ email, password }	{ access, refresh, user }	Returns JWT pair
POST	/api/auth/token/refresh/	Refresh token	{ refresh }	{ access, refresh }	Rotates tokens
POST	/api/auth/logout/	Bearer	{ refresh }	204 No Content	Blacklists refresh token

7.2 Complaints — Public

Method	Endpoint	Auth	Description
POST	/api/complaints/	None	Submit a new complaint. Rate limited to 5/hr per IP.
GET	/api/complaints/track/	None	Query: ?ref=CMP-2024-0001&email=x@y.com — returns status only.

7.3 Complaints — Admin

Method	Endpoint	Auth	Description
GET	/api/admin/complaints/	Admin JWT	List all complaints. Supports: ?status=, ?category=, ?search=, ?page=, ?ordering=
GET	/api/admin/complaints/{id}/	Admin JWT	Retrieve full detail of one complaint including notes and audit log.
PATCH	/api/admin/complaints/{id}/	Admin JWT	Update status only. Body: { status: 'in_progress' }
POST	/api/admin/complaints/{id}/notes/	Admin JWT	Add an internal note. Body: { note_text: '...' }
GET	/api/admin/complaints/stats/	Admin JWT	Returns counts by status for dashboard cards.

8. Frontend Screens & UI Specification

Screen	Route	Role	Key Ant Design Components	Description
Admin Login	/admin/login	Admin	Form, Input, Button, Alert	Email + password form. Error state shows Ant Alert. Redirect to dashboard on success.
Admin Dashboard	/admin/dashboard	Admin	Row/Col, Card, Table, Select, DatePicker, Input.Search, Tag, Pagination	4 stat cards at top. Below: filterable, searchable, sortable complaints table. Status shown as color-coded Tag.
Complaint Detail	/admin/complaints/:id	Admin	Descriptions, Steps, Tag, Timeline, Form, Button, Divider	Full complaint info in Descriptions component. Status update via Select + Save. Notes as Timeline. Audit log as collapsible Table.
Submit Complaint	/complaints/new	Public	Form, Input, Select, Button, Result	Multi-field form. On success: Result component with reference number and copy button.
Track Complaint	/complaints/track	Public	Form, Input, Steps, Tag, Alert	Ref number + email input. Shows Steps component reflecting current status progression.
404 / Unauthorised	*	All	Result	404 page for unknown routes. 403 for unauthorised access attempts.

8.2 Form Validation Rules (Frontend)

Field	Rule	Error Message
Full Name	Required, min 2 chars	Please enter your full name
Email	Required, valid email format	Please enter a valid email address
Title	Required, max 150 chars	Title must be between 1 and 150 characters
Description	Required, min 20 / max 2000 chars	Description must be at least 20 characters
Reference Number (track)	Required, pattern CMP-YYYY-NNNN	Please enter a valid reference number